Viola Charissa Diana Sembiring

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EDUCATION

University of Michigan, Ann Arbor **Ann Arbor, MI**
Bachelor of Science in Engineering (B.S.E) in Computer Engineering, Minor in Data Science *May 2028*
GPA: 3.94 / 4.00
Honors: William J. Brainstorm Prize (Top 5% of First-Year Students), University Honors and Dean's List (2024-2026).
Related Coursework: Programming and Data Structures, Discrete Mathematics, Multivariable Calculus, Differential Equations, Electronic Circuits, Logic Design, Signals and System, and General Physics.

Indonesia Maju Taman Sains Scholarship 3 **Jakarta, Indonesia**
Government-Funded College Scholarship and Pre-University Program *May 2024 - Present*

WORK EXPERIENCE

University of Michigan Electrical Engineering and Computer Science (EECS) **Ann Arbor, MI**
Instructional Aide, EECS 215: Introduction to Electronic Circuit *August 2026 – Present*

- Lead weekly lab sections of 20+ students, guiding circuit construction, measurement, and hardware debugging with oscilloscopes, function generators, and multimeters
- Grade weekly lab assignment, providing detailed feedback on circuit analysis and experimental results
- Hold weekly office hours for 370+ students in the course, clarifying lecture material and guiding them step by step through problem sets to strengthen their circuit analysis skills

PT. Teknologi Tujuh Asia **Jakarta, Indonesia**
Developer Intern *June 2026 – August 2026*

- Architected and launched a full-stack AI SaaS startup website end-to-end, leveraging LLMs to accelerate the entire design, development, and deployment process

University of Michigan Climate and Space Sciences and Engineering **Ann Arbor, MI**
Data Engineer Intern *May 2026 – August 2026*

- Applying dimensionality reduction techniques to multi-layered space physics datasets with 7 parameters to extract meaningful structure from high-dimensional sensor data
- Developing ML models to cluster, classify, and predict plasma and solar wind behavior, supporting ongoing research in space weather dynamics

Research Assistant (Advisor: Prof. Liang Zhao) *October 2025 – May 2026*

- Developed Python-based machine learning pipelines to preprocess and analyze large-scale in-situ solar wind data from SWICS Observatory and Parker Solar Probe (PSP) to support ongoing research on solar wind behavior
- Synthesized model outputs into technical visualizations and reports for senior researcher using Matplotlib and Seaborn, directly contributing to the interpretation of solar wind acceleration dynamics

Mathematics Learning Center (MLC) University of Michigan **Ann Arbor, MI**
Mathematics Undergraduate Tutor *August 2025 - Present*

- Facilitated individual and group tutoring sessions for 20–30+ students per week, simplifying complex concepts in Calculus I, II, and Differential Equations to improve student mastery and exam performance.

VOLUNTEER EXPERIENCE

Affirmation Pathway **Bali, Indonesia**
Co-Founder, Vice President, and Mathematics Mentor *March 2023 – April 2024*

- Helped 150+ students improve math skills through weekly tutoring sessions to underprivileged middle school students
- Led 30+ volunteer tutors by managing tutor scheduling and standardizing STEM teaching methodologies

TECHNICAL SKILLS

C | C++ | Python | MATLAB | Verilog | FPGA | Embedded Systems | Hardware Debugging | Linux | Git | Autodesk Fusion 360